

Project Title	Traffic Crash Analytics & Safety Intelligence Platform
Skills take away From This Project	Strong SQL querying, data analysis, database management, analytical thinking, Python-SQL integration, Streamlit table presentation, real-world data engineering exposure, business insight generation
Domain	Transportation Analytics

Problem Statement:

Traffic crash data contains valuable insights that can help improve road safety, optimize emergency response, and support policy decisions. However, extracting meaningful insights from large-scale structured data requires strong SQL and analytical skills.

In this project, provided with a **pre-cleaned and structured dataset** of traffic crashes. The objective is to:

- Analyze crash data using advanced SQL techniques
- Identify patterns, trends, and risk factors
- Generate business insights from structured data

The focus is on **SQL-based problem solving, analytical thinking, and insight generation**, similar to real-world data analyst and data engineer roles.

Business Use Cases:

1. Traffic Authorities

- Identify high-risk streets and accident-prone zones

2. Insurance Companies

- Analyze factors contributing to high-severity crashes

3. Emergency Services

- Evaluate response effectiveness using time-based data

4. Urban Planning

- Understand impact of road types and conditions

5. Research & Policy Making

- Study crash trends and contributing causes

Approach:

Phase 1: Data Loading & Understanding

- Load the provided cleaned dataset into SQL (SQLite/PostgreSQL)
 - Verify:
 - Row count
 - Schema
 - Data types
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Phase 2: Data Exploration

- Perform basic SQL queries:
 - COUNT, GROUP BY, ORDER BY
 - Understand:
 - Distribution of crashes
 - Common causes
 - Time-based trends
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Phase 3: Advanced SQL Analysis (Core Phase)

Learners must implement advanced queries using:

- Aggregations
- Window Functions (ROW_NUMBER, RANK, NTILE)
- CTEs (Common Table Expressions)
- Joins and subqueries

1. Find the **top 5 most dangerous combinations of weather and crash type** based on total crashes.
2. Identify the **top 10 streets with the highest number of injury crashes**.
3. Find the **percentage of crashes that resulted in injuries for each crash type**.
4. Determine the **peak crash hour for each month**.
5. Find the **top 5 primary causes of crashes during night time** (CRASH_HOUR $\geq$ 18).
6. Compare **average number of injuries in daylight vs darkness conditions**.
7. Find which **traffic control device type has the highest average injuries per crash**.
8. Identify the **top 5 locations (latitude/longitude) with the highest crash frequency**.
9. Find the **top 5 streets with the highest injury rate**, considering only streets with more than 100 crashes.
10. For each year, identify the **most common crash type**.
11. Find the **day of the week with the highest average crashes per hour**.
12. Identify **high-risk time slots**:
 - Group hours into buckets (Morning, Afternoon, Evening, Night)
 - Find which bucket has the highest injury crashes
13. Find the **top 3 contributing causes for each crash type**.

- (Use window functions like `ROW_NUMBER()` or `RANK()`)
 - 14. Calculate the **year-over-year growth rate of crashes**.
 - (Use `LAG()` window function)
 - 15. Identify **hotspot zones**:
 - Group nearby locations (round latitude & longitude to 2 decimal places)
 - Find top 10 zones with highest crashes
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Phase 4: Streamlit Integration (Mandatory)

- Build a **Streamlit application** to showcase SQL outputs
- All SQL query results must be:
 - Executed from the database (no hardcoded data)
 - Displayed in **table format using Streamlit (e.g., `st.dataframe()` or `st.table()`)**
- Each query should include:
 - A clear **title**
 - A **table output**
 - A short **business insight (1–2 lines)** explaining the result
- Optional:
 - Add filters (e.g., year, street, weather) using sidebar
 - Allow dynamic query execution based on user input

Results:

Learners should be able to:

- Write complex SQL queries confidently
- Identify:

- High-risk locations
- Peak crash times
- Dangerous conditions
- Generate **business insights from raw query outputs**
- Demonstrate strong **analytical thinking and SQL proficiency**

Project Evaluation metrics:

- Data loading accuracy and proper database structure
- Correct and efficient SQL query implementation
- Use of advanced SQL concepts (joins, CTEs, window functions)
- Analytical thinking and problem-solving approach
- Ability to generate meaningful business insights
- Integration of SQL queries into Streamlit
- Clear and interactive table display of results
- Code quality, readability, and best practices
- Overall completeness and correctness of the project

Technical Tags:

SQL, SQLite, PostgreSQL, Python, SQLAlchemy, Pandas, Streamlit, Data Analysis, Data Engineering, Window Functions, CTE, Joins, Analytical Queries

Data Set:

Data Set Link: [Traffic_CrashesData](#)

Property	Details
Dataset Name	Chicago Traffic Crashes
Source	Chicago Data Portal — data.cityofchicago.org
Format	CSV (comma-separated values)
Total Records	600,000+ rows (updated regularly)
Time Range	2015 to present
Spatial Coverage	City of Chicago, Illinois, USA
License	Public Domain — City of Chicago Open Data
Table Name (SQL)	CrashTable

Data Set Explanation:

- Each row represents a single crash event
- Dataset is pre-cleaned and standardized
- Includes:
 - Crash time and date
 - Location details

- Road and environmental conditions
- Injury severity data
- Contributing causes

Note:

- No major preprocessing required
- Dataset is ready for direct SQL analysis

Column Name	Type	Description
CRASH_RECORD_ID	TEXT	Unique identifier for each crash report
CRASH_DATE	TEXT	Date and time of the crash (MM/DD/YYYY HH:MM format)
POSTED_SPEED_LIMIT	BIGINT	Posted speed limit at the crash location (mph)
WEATHER_CONDITION	TEXT	Weather at time of crash (CLEAR, RAIN, SNOW, FOG, etc.)
LIGHTING_CONDITION	TEXT	Lighting condition (DAYLIGHT, DARKNESS, DARKNESS - LIGHTED, etc.)
FIRST_CRASH_TYPE	TEXT	Type of first collision (REAR END, SIDESWIPE, TURNING, etc.)
TRAFFICWAY_TYPE	TEXT	Road type (DIVIDED W/MEDIAN, NOT DIVIDED, ONE-WAY, etc.)

PRIM_CONTRIBUTORY_CAUSE	TEXT	Primary cause assigned by police officer
SEC_CONTRIBUTORY_CAUSE	TEXT	Secondary contributing cause
INJURIES_FATAL	DOUBLE	Number of fatalities in the crash
INJURIES_INCAPACITATING	DOUBLE	Number of incapacitating (serious) injuries
INJURIES_NON_INCAPACITATING	DOUBLE	Number of non-incapacitating injuries
INJURIES_TOTAL	DOUBLE	Total number of injuries of all types
CRASH_HOUR	BIGINT	Hour of crash (0–23)
CRASH_DAY_OF_WEEK	BIGINT	Day of week (1=Sunday through 7=Saturday)
CRASH_MONTH	BIGINT	Month of crash (1=January through 12=December)
BEAT_OF_OCCURRENCE	DOUBLE	Chicago Police beat where crash occurred
STREET_NAME	TEXT	Name of the street where crash occurred
LATITUDE / LONGITUDE	DOUBLE	GPS coordinates of the crash location

DATE_POLICE_NOTIFIED	TEXT	Date and time police were notified of the crash
year	TEXT	Derived year column from CRASH_DATE

Project Deliverables:

- Python source code files
- SQL database file (.db)
- SQL query results (screenshots or CSV)
- Streamlit app showing table outputs
- README file with setup steps
- Project report with insights 🚀

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Project Guidelines:

- Use **Python 3.9+** and write clean, readable code
- Follow proper **SQL standards** (uppercase keywords, avoid **SELECT ***)
- Use meaningful variable and function names
- Structure code into separate files (loading, analysis, app)
- Ensure all SQL queries run without errors and give correct results
- Display all query outputs in **Streamlit table format** with titles and insights
- Avoid hardcoding values; keep code reusable
- Add comments where necessary for better understanding
- Use **Git for version control** and commit changes regularly

- Maintain a clear project structure with proper documentation (README)

Timeline:

The project must be completed and submitted **within 14 days from the assigned date**

Reference:

Project Live Evaluation	 Project Live Evaluation
Capstone Explanation Guideline	 Capstone Explanation Guideline
GitHub Reference	 How to Use GitHub.pptx
Streamlit recording (English)	 Special session for STREAMLIT(11/...
PROJECT SERIES(English)	 Project Excellence Series: Guided L..
PROJECT SERIES(Tamil)	 Project Excellence Series: Guided L..

PROJECT DOUBT CLARIFICATION SESSION (PROJECT AND CLASS DOUBTS)

About Session: The Project Doubt Clarification Session is a helpful resource for resolving questions and concerns about projects and class topics. It provides support in understanding project requirements, addressing code issues, and clarifying class concepts. The session aims to enhance comprehension and provide guidance to overcome challenges effectively.

Timing: Monday-Saturday (3:30PM to 4:30PM)

Session Link:

https://docs.google.com/document/d/1J7z_Y2whgCtFDXixysXBNXqCuhXAYD9vkiNQ-7Q7XKs/edit?usp=sharing



LIVE EVALUATION SESSION (CAPSTONE AND FINAL PROJECT)

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About Session: The Live Evaluation Session for Capstone and Final Projects allows participants to showcase their projects and receive real-time feedback for improvement. It assesses project quality and provides an opportunity for discussion and evaluation.

Note: This form will Open only on Saturday (after 2 PM) and Sunday on Every Week

Timing:

Monday-Saturday (11:30AM to 1:00PM)

Monday-Saturday (05:30PM to 07:00PM)

Booking link : <https://forms.gle/1m2Gsro41fLtZurRA>

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